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Problem Statement

Ocean researchers, offshore industry, and government naval operations all require communication networks that can send data across the ocean under the water. However, this can be a challenge especially with higher bandwidth communications as simple radio frequency modems. Our communication network takes advantage of underwater acoustic (UW-A) and underwater optical (UW-O) modems to effectively output the most bandwidth depending on conditions.

Results

- Dynamic mode initially on the optical modem every 5 seconds (within range 50m). If out of range takes 1 second to readjust to acoustic modem to send data.
- Reliable signal quality using optical modem (low visibility, high UV light interference). Average throughput under test conditions measure 5.32 Kbits/sec. Although supporting theoretical rate (1 Mbps), environmental factors such as noise, alignment and packet overhead can reduce throughput.

Conclusions & Impact

We faced many challenges, such as the UW-A modem’s reliance on a TCP/IP socket whereas the UW-O relied on network made creating a program that can access both difficult. Resolving these differences, along with the testing of the modems in different conditions and creating the electrical and mechanical components were the bulk of the work. However, we were able to create a system that achieved our goal of an efficient communications unit that works under water.

This system can be applied to ocean reef research (this project is designed specifically for integration with another project, the KnifeBot). Otherwise, this technology can be used with military autonomous drones, live monitoring of underwater cables, or even deep-sea research missions with autonomous drones.

